

Statistical Equations Cheatsheet

Category	Equation	Description	Parameters	Notes
Mean (Average)	$\mu = \frac{1}{n} \sum_{i=1}^n x_i (\text{Population})$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i (\text{Sample})$	Measures the central tendency of a dataset.	x_i : Data points, n : Number of observations	Use sample mean for estimates.
Median	Middle value when data is ordered (for odd n); average of two middle values (for even n)	Central value separating higher and lower halves.	Ordered dataset	Robust to outliers.
Mode	Most frequently occurring value(s) in the dataset.	Indicates the peak of the data distribution.	Dataset	May be multimodal.
Variance (Population)	$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$	Measures the spread of data points around the mean.	x_i : Data points, μ : Mean, n : Size	Square of standard deviation.
Variance (Sample)	$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$	Unbiased estimate of population variance.	x_i : Data points, $\bar{x}$: Sample mean, n : Size	$n - 1$ for Bessel's correction.
Standard Deviation (Population)	$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$	Square root of variance, measures data spread.	x_i : Data points, μ : Mean, n : Size	In same units as data.
Standard Deviation (Sample)	$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$	Sample-based spread measure.	x_i : Data points, $\bar{x}$: Sample mean, n : Size	Used in inferential statistics.
Skewness	$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{s^3}$	Measures asymmetry of the distribution.	x_i : Data points, $\bar{x}$: Mean, s : Std Dev, n : Size	Positive (right-skewed), negative (left-skewed), 0 (symmetric).

Kurtosis	$g_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{s^4} - 3$	Measures the "tailedness" or peakedness.	x_i : Data points, $\bar{x}$: Mean, s : Std Dev, n : Size	Excess kurtosis; 0 for normal distribution.
Covariance	$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(y_i - \mu_y)$	Measures how two variables change together.	x_i, y_i : Data pairs, μ_x, μ_y : Means, n : Size	Positive (direct), negative (inverse).
Correlation Coefficient	$\rho = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$	Standardized measure of linear relationship.	$\text{Cov}(X, Y)$: Covariance, σ_x, σ_y : Std Devs	Ranges from -1 to 1.
Probability Mass Func. (PMF)	$P(X = x)$ = Probability of discrete outcome x	Probability mass for discrete random variables.	x : Outcome	Sum of PMF over all x = 1.
Probability Density Func. (PDF)	$f(x) = \frac{d}{dx} F(x)$	Density function for continuous variables.	$F(x)$: Cumulative distribution	Area under PDF = 1.
Cumulative Distribution Func. (CDF)	$F(x) = P(X \leq x)$	Probability that random variable $\leq x$.	x : Value	Ranges from 0 to 1.
Percentile Rank	Percentile Rank = $\left(\frac{\text{Number of values below } x}{\text{Total number of values } n} \right) \times 100$	Percentage of data points below a given value x .	x : Value, n : Total observations	Indicates relative standing.
Percentile Position	$\text{Position} = \frac{p}{100} \times (n + 1)$	Position to find the p^{th} percentile in ordered data.	p : Percentile (e.g., 25), n : Total observations	Interpolated if not an integer.
Interquartile Range (IQR)	$\text{IQR} = Q3 - Q1$	Measures spread of the middle 50% of data.	$Q1$: 25th percentile, $Q3$: 75th percentile	Used to detect outliers.

Additional Notes on Percentiles

- Percentile Value: Calculated by determining the position using

$$\text{Position} = \frac{p}{100} \times (n + 1)$$

and finding the corresponding data value. If the position is not an integer, linear interpolation between adjacent values is often applied.

- Quartiles : Specific percentiles ($Q1, Q2, Q3$) provide a quick summary of data distribution, with $Q2$ being the median.
- These measures are essential for understanding data distribution and identifying outliers when combined with IQR.